- Sequencing bias & errors
- Speed & RAM
- Data storage & visualization

# Specific challenges:
# (2) resequencing





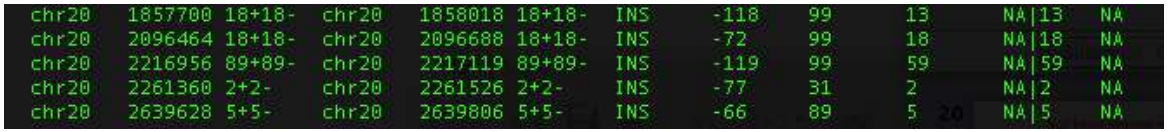
# Genetic variation detection

- what's structural variation (SV) ?
- why study SV ?
- how to detect SV ?

SNP




SV



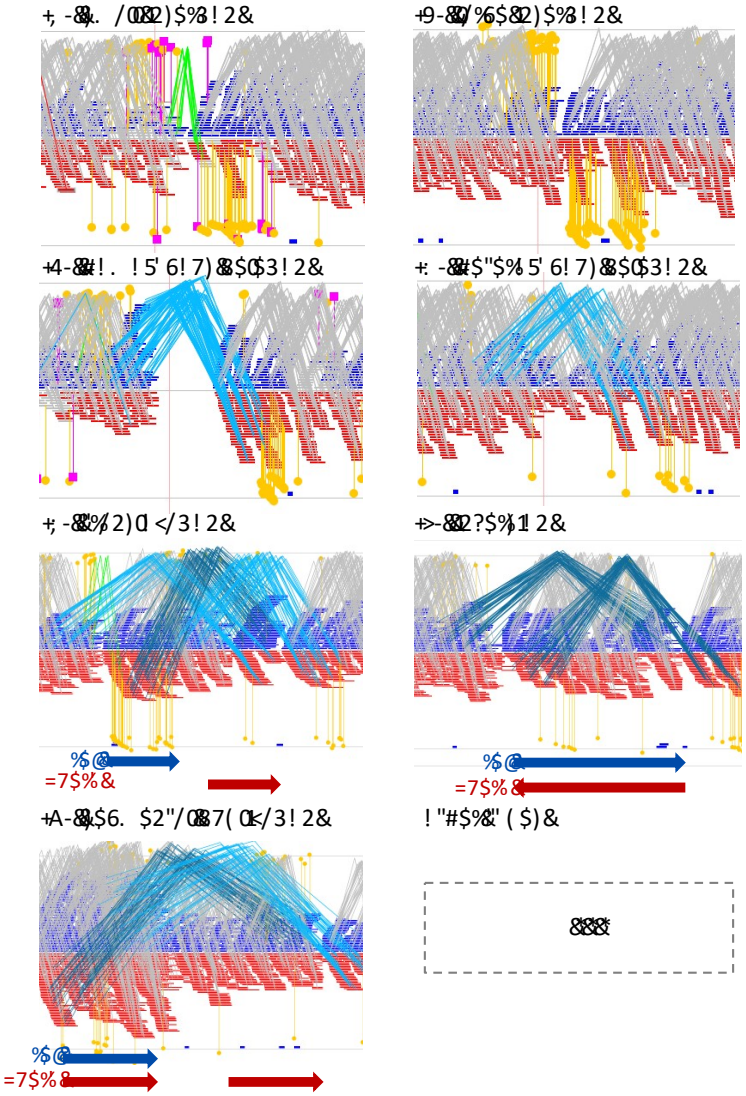

Challenges in detecting SVs：

- lack of intuitive visualization scheme；
- quality control；
- tandem repeats；
- complex forms of SVs；
- homo/hetero SVs。

## Most Genome Sequencing Projects Ignore SVs

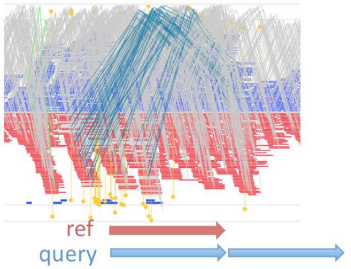
| Project | Technology | Paired End | SNPs; Short Indel | SVs | Reference |
|---|---|---|---|---|---|
| European-Venter | Sanger | Yes | 3M; 0.3M | 0.2M (>1000bp) | Levy et al., 2007 |
| European- Watson | 454 | No | 3M; 0.2M | Limited | Wheeler et al., 2008 |
| European- Quake | Helicos | No | 3M | Limited | Pushkarev et al., 2009 |
| Asian | Illumina | Partially | 3M; 0.1M | 2.7K (>100bp) | Wang et al., 2008 |
| HapMap Sample; Yoruban 18507 | Illumina | Yes | 4M; 10K | 0.1K | Bentley et al., 2008 |
| HapMap Sample; Yoruban 18507 | SOLiD | Partially | 4M; 0.2M | 5.5K (unknown definition) | McKernan et al., 2009 |
| Korean | Illumina | Yes | 3M | Limited | Ahn et al., 2009 |
| Korean- AK1 | Illumina | Yes | 3.45M; 0.17M | ~300 CNVs | Kim et al., 2009 |
| Three human genomes | Complete Genomics | Yes | 3.2-4.5M; 0.3-0.5M | Limited (50-90K block substitutions) | Drmanac et al., 2009 |
| AML genome & normal counterpart | Illumina | No | 3.8M; 0.7K | Limited | Ley et al., 2008 |
| AML genome | Illumina | Yes | 64 | Limited | Mardis et al., 2009 |
| Melanoma genome | Illumina | Yes | 32K;1K | 51 | Pleasance et al., 2009a |
| Lung cancer genome | SOLiD | Yes | 23K; 65 | 392 | Pleasance et al. 2009b |

# Basic forms of SVs

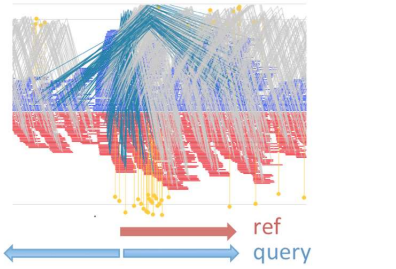


# Duplications

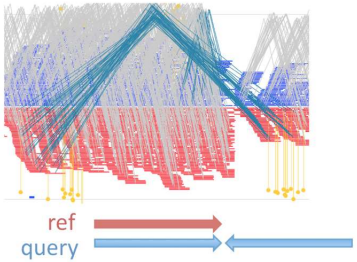
(A) tandem head-to-tail duplication



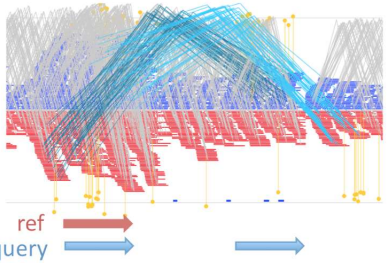

(B) tandem head-to-head duplication




(C) tandem tail-to-tail duplication




(D) segmental head-to-tail duplication




(E) segmental tail-to-tail duplication

